Notes on Selecting Attention Blocks

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Abstract

These notes collect the design decisions behind the block selection rule. They are written for readers who have already seen the main paper and want the details. We cover the choice of block size, the treatment of padding and the cost of the selection pass.

1 Block Size

The block size controls the trade off between the cost of selection and the granularity of the pattern. Small blocks approximate dense attention closely but make the selection matrix large. Large blocks make the selection cheap and force the kernel to read positions it does not need. We measured both effects on a single node with eight accelerators. A block size of sixty four was the best compromise on that hardware.

The optimal block size depends on the memory system rather than on the model. On a machine with a larger cache the optimum moved to one hundred and twenty eight. We therefore expose the block size as a configuration value rather than hard coding it.

2 Padding

Sequences rarely divide evenly into blocks, so the last block is padded. Padded positions are given a summary of zero and a score of negative infinity. This keeps the padded block out of the retained set without a separate branch in the kernel. The same trick is used for the causal mask inside a retained block.

The cost of the selection pass is the product of the number of query blocks and key blocks.

$$C = \frac{n^2}{b^2} \, d, \tag{1}$$

Here n is the sequence length, b is the block size and d is the head dimension.

3 Discussion

The selection rule is simple enough that it can be reimplemented in an afternoon. Most of the engineering effort went into the fused kernel rather than into the rule itself. We expect better summaries to matter more than a better selection rule [1]. A learned summary would add parameters, which we wanted to avoid for this release.

References

[1] M. Tanaka. Block structured kernels. In Proc. ICML, pages 991–1002, 2023.

[2] R. Roe. Efficient attention in practice. Journal of ML, 12(3):45–67, 2025.

Block size	Selection cost	Accuracy
16	12.4	88.9
64	1.9	88.7
256	0.4	85.1

Table 1: Cost of the selection pass in percent of a dense attention call, and accuracy on the held out split.

A Measurement Protocol

Every number in these notes is the mean of five runs. We discard the first run of each configuration because it includes kernel compilation. The machine was otherwise idle and the clocks were locked to their base frequency.